When a local pricing guard is not a welfare promise

A model-specific study of customer protection and learned pricing behavior

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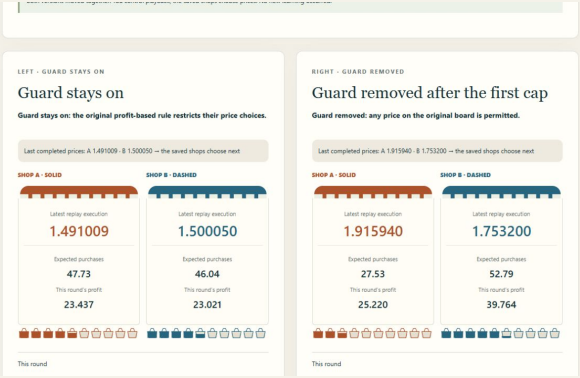
The Chinese University of Hong Kong (CUHK)

The pricing game

Two shops seek their own profit. Customers choose a shop or buy nothing.

The guard limits a proposed price using immediate profit against the rival's last price.

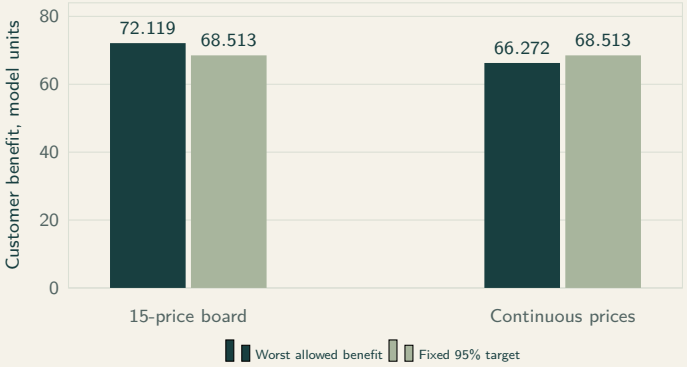
What customer benefit can it guarantee over many rounds?



Actual replay: both worlds show prices, expected purchases and shop profits.

Screenshot: saved policies after learning; manual Play and saved-policy Watch are separate. Customer benefit means model consumer surplus.

Price precision changes the guarantee



66.272

falls below the 68.513 target

The same initial cap. Separate finite and continuous calculations.

Report §3. Bounds shown rounded. These concern all permitted continuations, not observed learning.

Keeping the guard helps these learners



−12.733

Full grid after cap minus guard kept

All 12 favor keeping the guard

Same initial cap, clock, allocated memory and 2M-round budget.

One calibration and finite budget. More legal prices change learning difficulty. Mean and bars are descriptive; no confidence interval.

Repeating the cap adds further benefit



+0.956

Recurring cap minus startup guard

All 12 favor repeating the cap

Five clock labels and equal allocated memory in both arms.

Separate 24-run contrast. Every cycle-benefit endpoint changes between 1M and 2M rounds; no convergence claim.

Methods and verification

The shops use established tabular Q-learning

$$Q(s,a) \leftarrow Q(s,a) + 0.1\left[r + 0.95 \max_{a' \in \mathcal{A}(s')} Q(s',a') - Q(s,a)\right]$$

Exact saved-weight graph arithmetic measures the frozen repeating patterns.

Study	Runs	Rounds replayed	Checkpoints
Recurring cap	24	48 million	48
Guard removal	12	24 million	24

Separate implementations share V8 / Math.exp. The older 192-run study retains narrower replay coverage. These are verification counts, not extra samples.

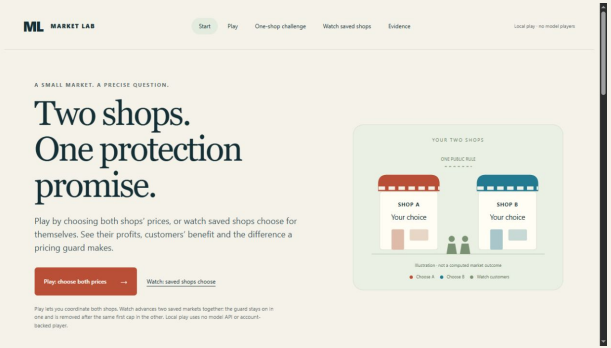
Contribution, limits and the game

Contribution

A controlled case linking a welfare guarantee to what pricing learners actually do.

Limits

One model and budget. One full-grid shop earns below static Nash. Dynamic incentives and continued adaptation remain open.



Play: choose both prices.

Watch: saved shops choose.

Local demo: `http://127.0.0.1:3003/`. Full methods, prior work, limitations and substantial AI assistance are disclosed in the report.